

BDG: Batch-Dispersion Guidance for Property-Band Generation with Continuous Flow Matching

A feedback-servoed deviation gain for property-gradient guidance, tested on QM9 3D molecules and on DNA enhancers on the probability simplex

Frame the study

Problem. Hit a property band around a target, without retraining the generator. A design campaign needs *many* usable candidates, so the quantity is **in-band**: the fraction of samples inside the band.

Question. Guidance concentrates a batch; that dispersion is a by-product of its strength. Does making dispersion a *requested* quantity raise coverage?

Hypothesis. At a centred target, coverage is limited by **batch spread**, not centring error — so a calibrated spread setpoint should raise it.

MODALITY 1

QM9 3D coordinates [1]

$\mathbb{R}^{N \times 3}$ with a relaxed atom-type simplex, $N \leq 29$.
Invariant to translation, rotation, permutation [2].

MODALITY 2

DNA enhancers on the simplex [3]

$(\Delta^3)^L$, one simplex per position over {A, C, G, T}. No rotation group; every position must stay on the simplex.

Two state spaces, not two datasets: a Euclidean coordinate space with a group acting on it, and a product of probability simplices.

Datasets, splits, and leakage

MODALITY 1 · QM9 [1]

133,885 molecules, explicit hydrogens, pinned by SHA-256

Split	Trains	n
train_a	generator and guide f_A	51,527
train_b	evaluator f_B alone	51,527
val	checkpoints, atom counts	17,748
test	held out	13,083

No leakage: the network that scores us never saw the data that trained the network that steers us. Normalisation comes from train_a alone; δ 's calibration set supplies neither targets nor atom counts.

MODALITY 2 · ENHANCERS [3]

6,126 Kenyon-cell *Drosophila* sequences

The evaluator is an **exact count**, so no split is needed and leakage is impossible.

We keep the two species apart. Their GC distributions differ, so steering spread on the mixture would partly steer the mixing proportion — the quantity under study.

Preprocessing (M1)

Coordinates zero-centred per molecule, in Å; dense [B, 29, ·] padding with a validity mask; properties in D / Bohr³ / Ha.

Representations, conditioning, handling

Representations

M1: coordinates plus a *relaxed* one-hot over {H, C, N, O, F}. The relaxation is what lets one continuous flow carry geometry and discrete types together.

M2: a point of $(\Delta^3)^L$. The simplex is continuous by the assignment's definition, and the native geometry of the sequence literature [4, 5].

Conditioning: none

Both base models are **unconditional**. The property enters only at sampling time, through guidance.

Deliberate: a conditional model gives the controller nothing to steer, and breaks the M1/M2 parallel we are testing.

Modality-specific handling

M1 invariance: translation by projection onto the zero-centre-of-mass subspace; rotation through squared pairwise distances, with updates along difference vectors; permutation by symmetric pooling. **12/12 numerical gates pass to $\sim 1e-17$.**

M2 mapping: sequences relaxed onto the simplex, decoded by per-position argmax, scored by an **exact** GC count.

Flow-matching baseline: the formulation

Data x_1 , source x_0 ; t runs 0 (noise) $\rightarrow$ 1 (data).

Probability path linear interpolant [6, 7, 8]	$x_t = t x_1 + (1 - t) x_0$	$u_t(x_t \mid x_0, x_1) = x_1 - x_0$
Source, not isotropic	Coordinates $x_0^c = \Pi_0 \epsilon$, with Π_0 the projection onto the zero-centre-of-mass subspace, dimension $3(N - 1)$. Type channels: a masked Gaussian.	
Loss, as implemented masked mean, two heads normalised separately, $\lambda = 1$	$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E} \left[\frac{1}{3 \sum M} \ M \odot (v_\theta^c - u^c)\ ^2 + \lambda \frac{1}{K \sum M} \ M \odot (v_\theta^f - u^f)\ ^2 \right]$ New symbols: M the atom mask, K the type channels, v_θ the learned velocity. Separate normalisation stops padding diluting the loss.	
Endpoint estimate closed form at every t	$\hat{x}_1 = x_t + (1 - t) v_\theta(x_t, t)$ The object every guidance arm differentiates.	

Flow matching: training, selection, sampling

Architecture [9]

Dense E(n)-equivariant graph network, width 256×8 layers, SiLU, **3,753,229 parameters**. Time enters at the input embedding. No attention.

Optimiser and budget

Adam $2e-4$, cosine to 5%, batch 256, clip 5.0, EMA 0.9999 with warm-up; sampling uses the EMA copy. **1500 epochs = 303k steps**, 11.4 h on one B200 MIG slice, seed 20260918.

Selection (pre-registered)

Highest *validation* atom stability at NFE 100, ties on loss, every 25 epochs.

Disclosed amendment. The rule picked epoch 1300; we ship **1500**, because a 512-sample signal cannot resolve the last 200 epochs. The 3-seed $\times$ 10,000 benchmark confirms it: +0.0018 atom, +0.0097 molecule, +0.0110 validity, same sign on all three seeds.

Sampling [6]

Probability-flow ODE, **explicit Euler**, uniform grid, **100 steps $\Rightarrow$ NFE 100**.

Quality saturates by 500 steps — 100 $\rightarrow$ 500 buys 0.65 atom and 2.8 molecule points — so 100 is a cheap operating point.

Why it matters for 2a: the closed-form endpoint estimate at every t is what makes this family convenient to guide.

Diffusion baseline: the formulation

Continuous-time variance-preserving process [2, 10, 11]; τ runs 0 (clean) $\rightarrow$ 1 (noise), β linear.

Noise schedule

$$\beta(\tau) = \beta_{\min} + \tau (\beta_{\max} - \beta_{\min}), \quad \beta_{\min} = 0.1, \quad \beta_{\max} = 20$$

Forward process

$$x_{\tau} = \alpha(\tau) x_0 + \sigma(\tau) \varepsilon, \quad \log \alpha(\tau) = -\frac{1}{2} \left(\beta_{\min} \tau + \frac{1}{2} (\beta_{\max} - \beta_{\min}) \tau^2 \right), \quad \alpha^2 + \sigma^2 = 1$$

Target: ε

The same two-term masked-mean loss as the flow model, with coordinate noise projected to zero centre of mass so the target is equivariant. Times $\tau \sim \text{U}(1\text{e-}3, 1)$ keep σ away from zero.

Reverse sampler

probability-flow ODE

$$\frac{dx}{d\tau} = -\frac{1}{2} \beta(\tau) (x + s_{\theta}(x, \tau)), \quad s_{\theta} = -\varepsilon_{\theta} / \sigma$$

Integrated backwards on a 100-step decreasing grid, then a jump to the Tweedie posterior mean.

What guidance consumes

$$\hat{x}_0 = (x_{\tau} - \sigma \varepsilon_{\theta}) / \alpha, \quad k = \sigma^2 / \alpha$$

Verified: $\alpha^2 + \sigma^2 = 1$ to 6 decimals, $\alpha(0) = 1$, $\alpha(1) = 0.0066$.

Diffusion: training, selection, sampling

Matched by construction. Same representation, split, EGNN backbone, optimiser, schedule, batch, EMA, clipping, seed and pre-registered selection rule as the flow model. It differs **only in the corruption process** — which is what makes 2a a comparison of families, not of engineering.

Architecture [9]

EGNN 256×8, 3.75M parameters, ϵ -prediction head. Adam 2e-4, cosine to 5%, batch 256, clip 5.0, EMA 0.9999, 1500 epochs.

Selection

Highest validation atom stability at NFE 100, ties on loss — the identical rule.

Sampling [2, 11]

100-step decreasing grid, then the Tweedie endpoint jump, so the unguided budget is **NFE 101** against the flow model's 100. We report the asymmetry rather than hide it.

Status, plainly. Our VP diffusion is implemented, gated and smoke-tested; the full-scale run is **[TODO: v3 Job A]**. Today's only *measured* diffusion comparator is a borrowed checkpoint, labelled as borrowed in 2a — so we claim no family winner from it.

How we will measure the two models

Both base models are ours, and **every number is scored by our own held-out evaluator f_B** — same network, same code, same band for both.

Quality and diversity

Atom and molecule stability, validity, uniqueness, novelty, diversity — EDM conventions [2], bond tables checked **entry for entry**. This decides which family we carry forward.

Control: in-band

Fraction with $|f_B(x) - y| \leq \delta$, where **$\delta = 2 \times f_B$'s calibration MAE** — pre-registered, so it cannot be tuned afterwards.

Decomposition

$RMSE^2 = bias^2 + spread^2$, in units of δ , plus spread relative to unguided. **Spread is what our innovation targets.**

Sample counts

Base models **3 seeds × 10,000**; guidance cells $n \times 3$ seeds, $n =$ **[TODO: v3 n]**; screening cells 256–512, always labelled.

The limit of in-band. It rewards *any* arm that contracts the batch, including onto implausible structures — so every in-band number appears beside the chemistry it spent.

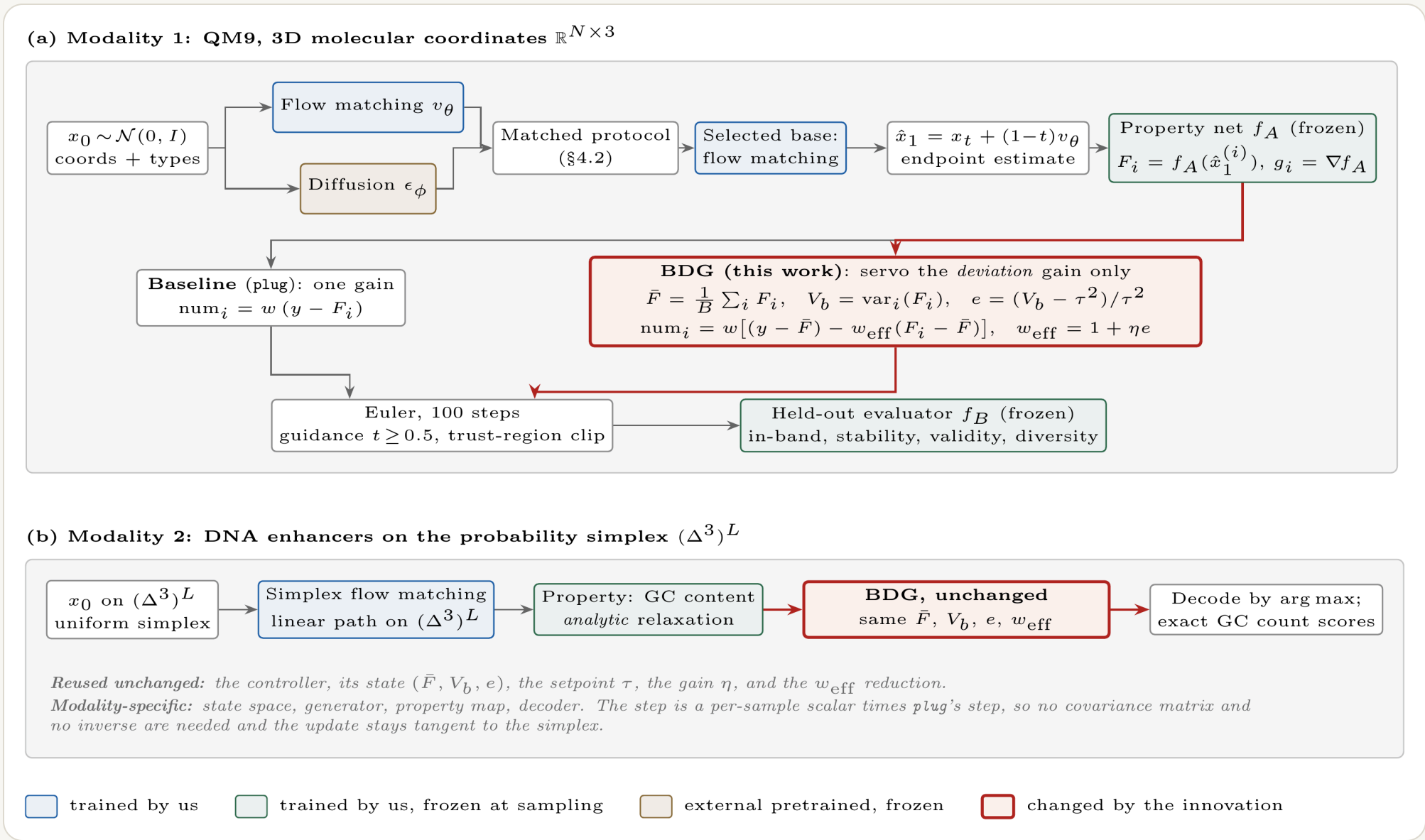
Flow matching vs diffusion: a matched protocol

Both models are ours, trained on the same half of QM9. We hold **everything but the corruption process** fixed, so 2a compares two *families* — not two engineering efforts.

Factor	Identical by construction	Necessarily different
Data and split	train_a , 51,527 molecules; same representation	—
Backbone	EGNN 256 × 8, 3.75M parameters each [9]	velocity head vs ϵ head
Corruption	—	linear interpolant vs variance-preserving — the one thing under test
Training recipe	Adam 2e-4 cosine, batch 256, clip 5.0, EMA 0.9999, seed 20260918, 1500 epochs	completed run pending [TODO: v3 Job A]
Selection	highest validation atom stability at NFE 100, ties on loss	—
Sampling	100-step probability-flow ODE	diffusion's Tweedie jump: NFE 101 vs 100
Evaluation	our f_A and f_B, one δ , one metric implementation	—

Paired, then tested. Arms share starting noise, seeds and atom counts, so we compare molecule by molecule; binomial standard errors over three seeds, with multiplicity-corrected significance beside the single-test value.

Study overview: how the pieces fit



PANEL (A), LEFT TO RIGHT

- 1 Two families, one representation · 1c · 1d**
The same molecules and features train a flow-matching [6] and a diffusion [2] model.
 - 2 A matched protocol picks one · 1e · 2a**
Identical recipe and metrics, so the choice is about families.
 - 3 Guidance reads a frozen predictor · 2b**
 f_A is differentiated at the endpoint estimate, then pulled back.
 - 4 BDG changes one gain · 2c**
Centring stays at w ; only the deviation gain is servoed.
 - 5 A separate network scores · 2b**
 f_B , from the disjoint half, never enters the sampler.
- b Panel (b): same controller, new state space · 3a**
Generator, property map and decoder change [4]. The controller does not.

Colour: blue trained by us · green frozen at sampling · tan external pretrained · **red changed by the innovation.**

Choose the base model

Every row is unconditional and unguided — no target, no property gradient. Guidance starts in 2b.

Model	Trained on	Atom stab. ↑	Mol. stab. ↑	Validity ↑	Uniq. ↑	NFE ↓
Flow matching (ours)	train_a (51,527)	0.9366 ±.0012	0.3993 ±.0085	0.7625 ±.0031	0.9939	100
VP diffusion (ours)	train_a (51,527)	[TODO: v3 Job A]	[TODO]	[TODO]	[TODO]	101
EquiFM [25], borrowed	full QM9 (100K)	0.9837	0.8672	0.9453	1.0000	100
Real QM9 — the ceiling	—	0.9940	0.9560	0.9820	1.0000	—

§4.2. Euler probability-flow ODE, our evaluator, EDM conventions [2]. Ours: 3 seeds × 10,000. EquiFM: our own 256-molecule pilot on the released checkpoint, one seed, no error bar — and it saw twice our training data (appendix A1).

Flow matching goes forward — not for a fidelity win

The row that would settle the family question is still running, and EquiFM cannot stand in for it: external model, twice the data, different probability path.

The reason is structural — flow matching's linear path gives a **closed-form endpoint estimate at every t**, the object every guidance rule differentiates.

Why we sit below the ceiling

1 • Unscaled one-hot features — the published ablation removing exactly that scaling costs 3 atom and 35 molecule points [2].

2 • Training budget — 303k steps against ~1.56M.
Ruled out: the half-data split (~0.3 points [26]) and our evaluator.

Guidance strategy

Rule and target

$$G_i = \frac{\text{num}_i}{s^2} J_i^\top g_i, \quad \text{num}_i^{\text{plug}} = y - F_i, \quad F_i = f_A(m_i), \quad g_i = \nabla_m f_A(m_i)$$

The frozen guide f_A is differentiated at the endpoint estimate m_i , and the gradient is pulled back to the sampling variable. The target y is each property's **q50**. This is a DPS-style plug-in Gaussian-energy field [12].

Nothing is trained at guidance time

Generator v_θ — ours, frozen at sampling.

Guide f_A — ours on `train_a`; frozen, gradients disabled.

Evaluator f_B — `train_b`; frozen, and never inside the sampler.

Strength, normalisation, safety cap

Strength: the correction is scaled by $w \cdot (1 - t)/t$ [12, 14];

normalisation: by s^2 , the guide's property standard deviation.

Safety cap (trust region): each correction is shrunk until it is no longer than the model's own velocity, $\|G_i\| \leq \|v_i\|$ — prior practice [13], identical for every arm.

Window: guidance is on for $t \geq 0.5$ only.

The cap is doing real work. Uncapped, one correction can overwhelm the velocity, the trajectory diverges and the coordinates overflow to NaN: the same diagnostic run gave **222 broken molecules** without the cap and **zero** with it.

Guidance works, and what it costs

Arm	in-band ↑	Spread / unguided	Mol. stab. ↑
unguided	.090 / .121 / .051	1.00 / 1.00 / 1.00	.434 / .434 / .434
plug-in (best w) [12]	.184 / .219 / .133	0.52 / 0.60 / 0.47	.246 / .199 / .320
TFG (w = 4, μ only) [15]	.535 / – / –	0.22 / – / –	.172 / – / –
BDG (w = 32, τ_mult = 0.5)	.180 / .227 / .106	0.49 / 0.68 / 0.53	.348 / .262 / .340

§4.3. Cells are μ / gap / α at q50, otherwise matched: one generator, one seed, n = 256, best strength per arm over w ∈ {0.05, ..., 32}, no chemistry floor. Three-seed confirmation: **[TODO: v3]**

The target improves

Plug-in: **2.0×** μ, **1.9×** gap, **2.6×** α over unguided. TFG reaches **6.0×** on μ.

The cost is real

Molecule stability falls 0.434 → 0.246 (plug-in, μ) and → 0.172 (TFG). Validity follows.

The mechanism is visible

Spread contracts 1.00 → 0.22–0.60. Guidance works **by concentrating the batch**: the arm that contracts most wins in-band and loses the most chemistry.

That trade-off is inherited, not chosen. 2c asks whether the user can choose it instead.

Motivating the innovation

1–2

band widths off target
centring error

5–9

band widths of spread
what actually binds

The weakness. At a centred target, coverage is a *dispersion* problem. Yet every guidance rule optimises a *centring* objective through **one dial, w** , which moves centring and spread together in a ratio the user never chooses.

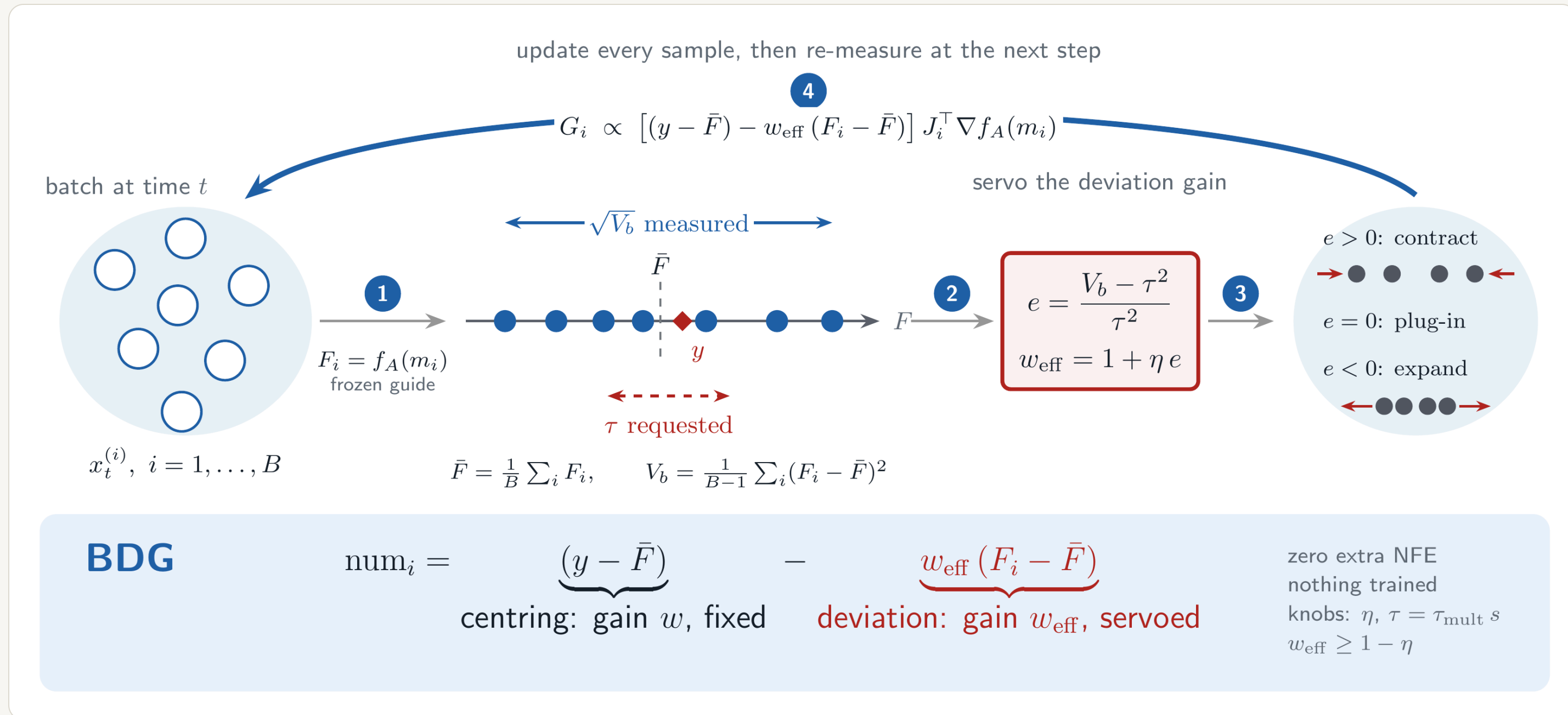
The hypothesis. Let the sampler *measure* its own batch dispersion and servo toward a *requested* value: the user then trades spread against centring deliberately, in property units rather than units of push.

Prior art	Already owns
CFG-Ctrl [16], Feedback Guidance [17]	the guidance scale read as a control gain
MGD [18]	ensemble moments driven to a target, two-sided
Variance-Tilted Diffusion [19]	batch-spread targeting; widen-only, fixed feature
Particle Guidance [20]	batch-coupled repulsion, no setpoint
Negative-guidance window [21]	negative weight widens the distribution

We claim none of these ingredients

A bounded search (~8 queries, abstracts only, 25 Sep 2026) did not turn up a scalar weight set by the batch variance of a *learned predictor's own endpoint prediction*, against a *two-sided user-chosen setpoint, signed through zero*. That bounds our claim; it is not a priority claim.

BDG: the mechanism



A descent direction on what binds. $\partial V_b / \partial F_i = 2(F_i - \bar{F}) / (B - 1)$, carried to m_i along g_i . The $2 / (B - 1)$ is absorbed into η , so the gain does not scale with B .

Free, and frozen. Both terms multiply the same g_i , so BDG shares plug-in's single backward pass [12]. Generator and f_A untouched; two new knobs, η and τ .

BDG: the reduction, and its bounds

We state the reduction ourselves: num_i is affine in F_i , so it factors exactly

$$(y - F_i) - \eta e (F_i - \bar{F}) = (1 + \eta e) (y_{\text{eff}} - F_i), \quad y_{\text{eff}} = \frac{y + \eta e \bar{F}}{1 + \eta e}$$

Verified to **3.0e-15** analytically, **6.5e-7** on the real field.

So what is BDG?

At any instant, **the plug-in field at a rescaled weight aiming at a shifted target**. No new direction: any guidance affine in F_i is the plug-in rule reweighted.

What survives it

w_{eff} is set **online** from a measured quantity, and may **change sign** — which w cannot do without reversing the centring term too.

Bound, for any batch

$$V_b \geq 0 \Rightarrow e \geq -1 \Rightarrow w_{\text{eff}} \geq 1 - \eta$$

This licenses running two-sided with no clamp. **Design constraint $\eta > 1$** , or w_{eff} never goes negative and the controller cannot expand.

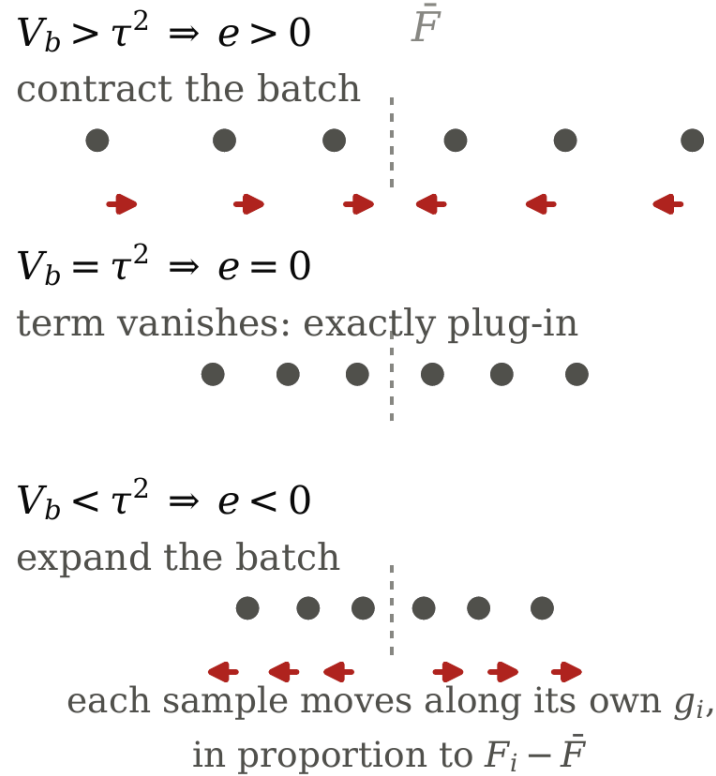
Equilibrium, honestly

$$w_{\text{eff}} = 0 \Rightarrow V^* = \tau^2 (1 - 1/\eta)$$

An **asymptote the 50-step window never reaches**. So we never say the controller holds its setpoint, and we order every table by *measured* w_{eff} .

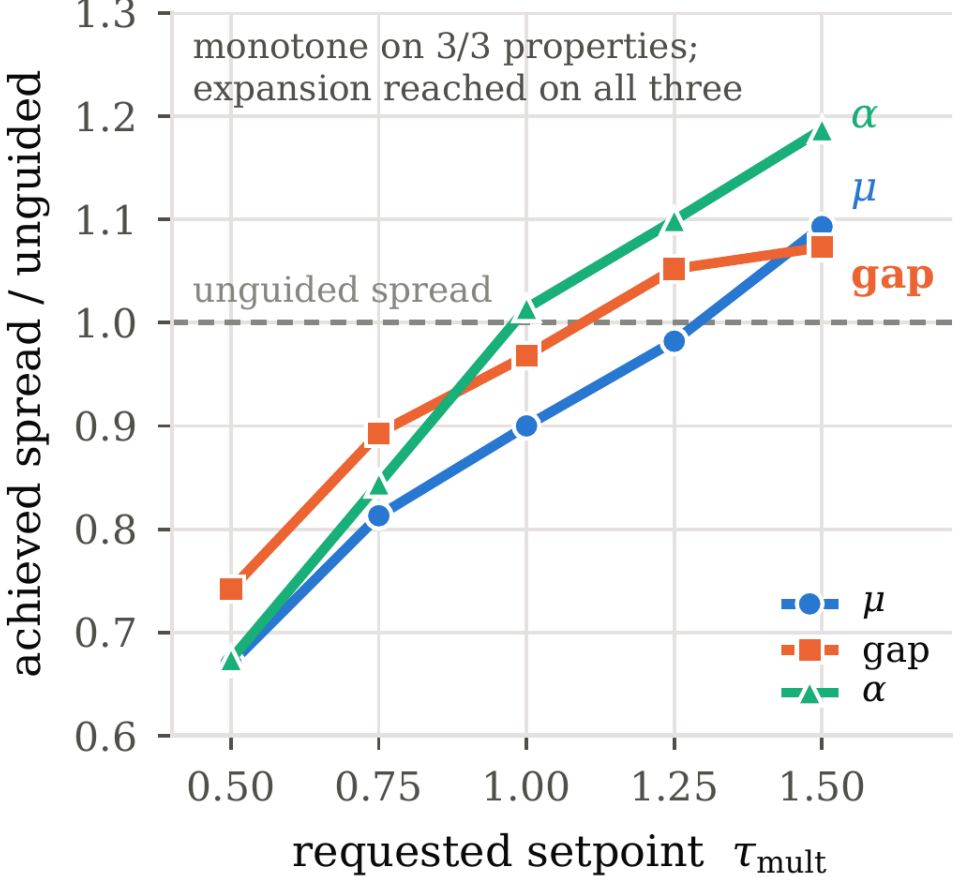
The controller, measured

(a) the sign mechanism



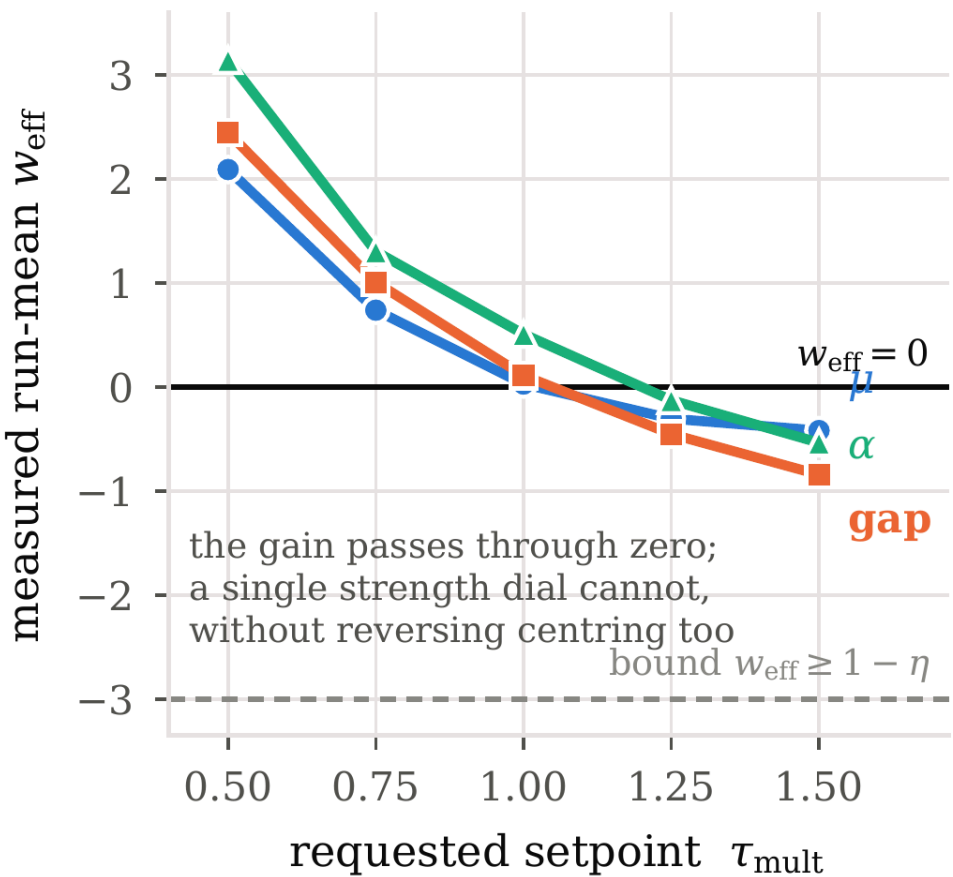
(a) **Sign is the mechanism.** $e = 0$ recovers the plug-in rule exactly.

(b) setpoint controls achieved spread



(b) **The knob works.** Setpoint $\rightarrow$ achieved spread is **monotone on 3/3** properties, and crosses *above* the unguided spread on all three — which the strength dial does not reach.

(c) the deviation gain changes sign



(c) **The gain passes through zero**, inside its bound. A single strength dial cannot do that without reversing the centring term as well.

Testing the innovation: ablations

§4.4. BDG at q50, ordered by *measured* w_{eff} , never by τ_{mult} . Port cells: $n = 256$, one seed, $w = 1$, own unguided control.

Variant	Role	η	τ_{mult}	Spread / ung.	in-band ↑	Mol. stab. ↑
plug-in, $w = 1$	base	—	—	0.714	0.1094	0.3633
BDG $\eta = 0$	gate	0	1.0	0.714	0.1094	0.3633
BDG one-sided, asked to expand	gate	4	1.5	0.714	0.1094	0.3633
BDG contract	ladder	4	0.5	0.670	0.1406	0.3359
BDG neutral	ladder	4	1.0	0.900	0.0820	0.4258
BDG expand	ladder	4	1.5	1.093	0.0703	0.4141
plug-in, $w = 4$	control	—	—	0.584	0.1406	0.2656

Gates prove the code, not the claim
 $\eta = 0$ is **bit-identical** to plug-in: the dispersion term is the only addition.
One-sided asked to expand is bit-identical too (raw $e = -0.984$ clamped to 0), so the sign branch is real and the clamp self-consistently inert.

Control · open-loop e replay
Fresh noise, $\eta = 4$, $\tau_{\text{mult}} = 0.5$: recovers **69.5–96%** of the closed loop.

Control · signed-w plug-in
[TODO: not run; drop rule applies]

Three-seed v3 grid
[TODO: $\eta \in \{0,1,2,4,8\} \times \tau_{\text{mult}} \in \{0.5,0.75,1,1.5\}$, $w \in \{1,4\}$]

What the ablations establish

Comparison	Rules out
$\eta = 0$ vs plug-in	that anything but the dispersion term changed
one-sided vs two-sided	that the expansion branch is decorative
the τ ladder	that the setpoint does not reach achieved spread
plug-in at $w = 4$	that coverage came from controlling, not pushing
open-loop e replay	that the effect is feedback, not a time-varying schedule
signed-w plug-in [TODO]	that the same spread is not available more cheaply

1 · The setpoint controls spread

Monotone on **6 of 6** curves: 0.670 $\rightarrow$ 1.219 on the guide's scale, 0.650 $\rightarrow$ 1.197 on the evaluator's, reaching expansion above 1.0 on all six. w_{eff} changes sign up to **34 times** in the 50-step window, spanning $[-2.64, +10.35]$ against the bound ≥ -3 .

2 · Coverage does not improve

No rung beats plug-in at $|z| \geq 3$: over **72 paired tests** the largest is 3.07, which Šidák turns into **p = 0.14**. Removing the chemistry floor does not change it.

3 · The trade-off moves instead

At *matched* in-band 0.1406, BDG keeps molecule stability **0.3359** against plug-in's **0.2656**.

Honest reading. The schedule largely transfers across seeds (69.5–96% recovered open-loop); feedback adds a small measurable remainder (0.3–6.6%). We do not claim the loop is irreducible to a schedule.

Position against recent work

Method	in-band ↑	Mol. stab. ↑	Diversity ↑	Gen. passes	Guide passes
DPS-style plug-in [12]	[TODO: v3]	[TODO]	[TODO]	10,000	4,000
TMPD-inspired [22]	[TODO: v3]	[TODO]	[TODO]	10,000	4,000
LGD-MC [23]	[TODO: v3]	[TODO]	[TODO]	12,000	20,000
TFG [15]	[TODO: v3]	[TODO]	[TODO]	8,000	20,000
Flow matching, unguided	[TODO: v3]	[TODO]	[TODO]	10,000	0
BDG (ours)	[TODO: v3]	[TODO]	[TODO]	10,000	4,000

§4.5. Every row is reproduced inside our pipeline, on our frozen checkpoint, guide and evaluator, at q50 and w = 1. Passes are per cell at n = 5,000.

Why these four

All are inference-time guidance from the last five years, all steer a frozen generator with a frozen predictor, and all run on *our* checkpoint — which isolates **the guidance rule**, the thing our contribution is about.

Efficiency, measured

BDG's two terms share one backward pass: **4,000** guide passes per cell against **20,000** for LGD-MC and TFG. Not the cheapest overall — TFG uses fewer generator passes.

Fair study, and fair discussion

Reproduced, not quoted

Every 2e row is **reproduced by us** on one frozen checkpoint, guide, evaluator, target and sample count. The only quoted numbers in the study are 2a's published base-model rows, labelled and never pooled with ours.

Three port caveats

Plug-in is a DPS-*style* Gaussian-energy field; the original differentiates the residual L2 norm [12].

TMPD is the local-linear scalar limit: a related-work comparison, not a faithful nonlinear port [22].

TFG runs at its published QM9 hyperparameters, but through our deterministic ODE sampler, and our unscaled one-hot features move $\sim 16\times$ less relative to coordinates than the released checkpoint's.

Never read these against TFG's published table [15].

Better

Chemistry retained at matched coverage. A fifth of LGD-MC's and TFG's guide passes.

Comparable

in-band against the plug-in rule — the honest reading of the ablation null.

Worse

TFG reaches far higher in-band (0.535 vs 0.180 on μ) at a chemistry cost we judge unacceptable (0.172 vs 0.348), and uses fewer generator passes.

[TODO: v3 completes this comparison]

The transferred method

Unchanged: the whole controller

$\bar{F}$, V_b , e , the setpoint τ , the gain η , the bound $w_{\text{eff}} \geq 1 - \eta$, the reduction. **Zero code change.**

It can transfer because BDG's step is a per-sample scalar times the plug-in step: it inherits that step's tangency and needs **no covariance matrix and no inverse**. That matters here — the natural simplex covariance $\text{diag}(p) - pp^T$ is **singular**.

Changed	M1 (QM9)	M2 (DNA) [4, 5]
State space	$\mathbb{R}^{N \times 3}$	$(\Delta^3)^L$
Generator	EGNN	linear-path flow on the simplex
Property	trained equivariant net	analytic differentiable GC content
Evaluator	f_B	exact GC count
Decoder	none	per-position argmax

Swapping a learned property for an exact function is the point: the transfer tests the **controller**, not the property network.

Training (M2)

Linear-path flow matching on $(\Delta^3)^L$ over all 6,126 sequences [3], validation-based early stopping, checkpoint on validation loss. Architecture and budget: **[TODO: v3 Job B]**.

Sampling (M2)

Same solver family, step count, window and velocity-relative trust region as M1; decode by argmax. The clip fired on **0–3%** of steps; the expanding cell clipped **once in 25,600**.

Transfer results

Structural signature	QM9	DNA simplex
$\eta = 0$ reproduces plug-in	bit-identical	bit-identical
one-sided at an expanding setpoint	exact	exact, $w_{\text{eff}} = +1.000$
setpoint ladder monotone in spread	0.67 $\rightarrow$ 1.22	0.81 $\rightarrow$ 1.01
expansion reached	1.168	1.006
cost identical to plug-in	yes	yes, byte-identical
contraction breaks the fidelity floor	yes	yes
in-band improves	no	no

vs recent simplex models	Dirichlet FM [4]	Fisher Flow [5]	MOG-DFM [3]	BDG (ours)
GC in-band, DNA simplex	[TODO: v3 Job B]	[TODO: v3 Job B]	[TODO: v3 Job B]	[TODO: v3 Job B]

Fair discussion. Every structural signature transfers; coverage improves on neither modality; the external comparison is **[TODO: v3 Job B]**.

Not the published benchmark

Standard: class-conditional generation at **500 bp** on 81-/47-class sets of ~90,000 sequences, scored by FBD over 10,000 samples. Ours: GC-band coverage, **6,126** sequences, **200 bp**. Our base generator is the linear-FM baseline those papers report, but the numbers are not comparable and are never pooled.

Further disclosures

Gumbel-Softmax FM [24] never runs the enhancer benchmark, so only two of the three closest named methods apply.

Fisher Flow [5] disputes the standard evaluator: its cell-type classifiers score 11.5% / 11.2% test accuracy.

MOG-DFM's guided demo runs at length 100 while reporting base FBD at 500 bp [3].

Does the innovation transfer?

DIRECTION

Yes

on every signature we can check.

Gates hold exactly, the ladder is monotone, expansion is reachable, cost is unchanged — and **the null transfers too**: spread control raises coverage in neither state space, and in both it is the *contraction* rung that breaks the fidelity floor.

MAGNITUDE

2.6× smaller

0.20 against 0.52 in achieved spread. We attribute it to the property map, not the controller, and measured all three causes:

GC is affine in the simplex coordinates — identical gradient at every position, zero curvature — and renormalisation partly undoes the edit.

GC is a 200-position average, which suppresses the very variance the controller reads.

The band spans only 4.3 attainable GC values at granularity 1/200 — a ceiling on what any method could move.

Ruled out: the trust region (fired on 0–3% of steps; the expanding cell clipped once in 25,600).

The evidence supports transfer of the **mechanism and its limits**, and locates the magnitude gap in a measurable property of the task. The lever for a larger effect is a **richer property** — a trained cell-type classifier logit used as classifier guidance rather than conditioning.

Limitations and failure modes

1 · The clip is a live confound at large $|w_{\text{eff}}|$

Plug-in clips on **884** sample steps; the contracting rung on **2,516**. At w_{eff} of order 10^3 the dispersion term sits far outside the trust region, so what reaches the sample is the *clip's truncation* of it. A rung can be **clip-limited** rather than controller-limited, and no metric here separates the two. **Fix:** a matched-clip-activity run.

2 · The trade-off is re-parameterised, not removed

Chemistry cost tracks $|w_{\text{eff}}|$, not the *direction* of the knob: strongly contracting and strongly expanding rungs both cost chemistry. A descriptive correlation over coupled cells, not a test.

3 · The batch is the estimator

V_b is computed over whatever tensor the sampler is handed, so n samples in several batches run *several independent controllers*. Our headline batch of 500 is a memory constraint, not a modelling choice, and we disclose it wherever n exceeds it.

4 · The flow drifts on its own

Across-molecule variance *contracts and then re-expands* on **unguided** runs, so a setpoint on V_b is chasing a moving quantity.

Two of these are **measured confounds**, not speculation — which is why the controller claim is provisional in 3e.

Conclusion and future directions

Does a requested dispersion raise band coverage?

No.

No rung beats plug-in at a corrected significance level.
But we can say why.

What the two modalities do support

Servoing the deviation gain gives **monotone, two-sided, modality-portable** control of batch spread at **no extra sampling cost**.

It does not remove the coverage-against-fidelity trade-off. It makes that trade-off **a quantity the user requests**.

Why chemistry survives

The coefficient is affine in F_i , so BDG stays inside the plug-in field's reachable set.

Holding centring at w and servoing only the deviation gain buys contraction **from one mode instead of the whole field** — which is why molecules survive at matched coverage.

The experiment that could withdraw the claim

The **signed- w plug-in**, under our pre-registered drop rule: if it reaches BDG's spread at equal chemistry, we drop the controller claim and report the decomposition instead.

Then

Matched clip activity — separate clip-limited from controller-limited rungs. **A second seed pair** for the open-loop replay. **A richer M2 property**, nonlinear and motif-driven.

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APPENDIX · NOT PART OF THE 15-MINUTE TALK

Q&A backup

Four topics, one or two slides each.

A1

EquiFM, the borrowed generator

Architecture, path, and how our arms run on it

A2

Property predictors

Ours vs TFG's: architecture, training, accuracy

A3

The data split

Who trains on which half, and why halves

A4

The guidance arms

Each arm's correction, what it changes, what it costs

EquiFM: the borrowed generator

Released checkpoint	
Paper	Song et al., NeurIPS 2023 [25]
Backbone	EDM-style equivariant GNN dynamics [2, 9], 256×9
Parameters	5,339,920 (EMA)
State per atom	3 coordinates (Å) + 5 one-hot ÷ 4 + charge ÷ 10; explicit H
Training data	the full 100K QM9 partition, unconditional

Hybrid probability path

Native time τ : 1 = noise, 0 = data.

Coordinates: almost linear — but the noise is rotated (Kabsch) and permuted (Hungarian) onto the molecule first, an equivariant optimal-transport coupling, plus an angle penalty.

Types and charge: variance-preserving, β from 0.1 to 20.

Sampling: Euler, $\tau = 1 \rightarrow 0$.

$$x_\tau = (1 - \tau) x_0 + s_x(\tau) z, \quad s_x = \epsilon + (1 - \epsilon)\tau$$

$$h_\tau = a h_0 + \sqrt{1 - a^2} z', \quad a = e^{-T(\tau)/2}$$

Our unguided pilot

256 molecules, 100 Euler steps: atom **98.37%**, molecule **86.72%**, validity **94.53%**, uniqueness 100%. A usability check, not a benchmark.

Why its chemistry beats ours

Four candidates, none isolated by us: $\sim 2\times$ the training data, 5.34M vs 3.75M parameters, one-hot scaled by $\frac{1}{4}$ (our leading suspect in 2a), and the aligned coordinate path.

Running our guidance arms on EquiFM

Why it is in the study: a second, external flow-matching generator asks whether the comparison survives on a model we did not train. Backend **equifm**, all seven arms [25].

Endpoint estimate m	$m_x = (1 - \epsilon) x_\tau - s_x q_x, \quad m_h = a h_\tau + q_h/a$	Exact — algebra from its own regression targets.
Covariance Σ TMPD and LGD-MC only	$\Sigma = KJ, \quad k_x = \frac{s_x^2}{1 - \tau}, \quad k_h = \frac{1 - a^2}{a}$	Exact for types, approximate for coordinates , whose aligned noise is not an independent Gaussian channel.
Guided velocity	$\Delta v_x = -\frac{s_x}{1 - \tau} G_x, \quad \Delta v_h \approx -\frac{\beta(\tau)}{2} G_h$	The model's own velocity map at the guided mean $m + K \cdot G$.

Property pair here

TFG's released guide (tf_predict at time 0) and oracle (evaluate) [15]. Its band is narrower than ours — δ 0.157 vs 0.175 (μ), 0.165 vs 0.506 (α), 0.0037 vs 0.0075 (gap) — so **in-band never crosses backends**.

Disclosed limits

Trained on the full 100K, so it saw both halves. Only TMPD and LGD-MC touch the Σ surrogate; plug-in, TFG and BDG do not. The charge channel is held fixed during guidance. 1.85× our memory per molecule.

Property predictors: our pair and TFG's

Ours score the **fm** and **EDMsecond** backends; TFG's score **EquiFM**. f_A's **gradient** steers sampling; f_B only **scores**.

	Ours — f_A and f_B	TFG guide (tf_predict)	TFG oracle (evaluate)
Network	invariant EGNN on squared distances, mean-pool, linear head [9]	EDM dynamics EGNN with an energy head [2, 15]	EDM property-classifier EGNN [2]
Width × layers	128 × 4	192 × 7	128 × 7
Edge attention	no — equal-weight sum	yes	yes
Parameters (μ)	429,825 each	1,671,566	743,944
Inputs	coordinates + unscaled one-hot	+ one-hot ÷ 8; time fixed at 0	+ nuclear-charge powers
Training data	f_A: train_a · f_B: train_b (51,527 each)	QM9 second half (~50K)	QM9 first half (~50K)
Recipe	clean molecules, MSE on a standardised target; Adam 3e-4 cosine, batch 128, 120 epochs (~23 min); best validation MAE	clean molecules; EMA at epoch 2,000	best-validation checkpoint

Theirs are better; ours are disjoint

MAE	μ (D)	α (Bohr ³)	gap (Ha)
our f_A	0.0872	0.2325	0.00402
our f_B	0.0877	0.2531	0.00374
TFG guide	0.0417	0.0925	0.00185
TFG oracle	0.0784	0.0824	0.00187

Same 3,000 calibration molecules. Ours are scored only on the half each did not train on; TFG's on the whole pool, which overlaps their training halves and flatters them.

The honest gap on μ . On molecules neither TFG net trained on: guide 0.0659 D, oracle 0.0728 D, our f_B 0.0840 D. **~20% better, not 2×.**

Why theirs are more accurate
Capacity: 2–4× the parameters, 7 layers not 4.
Attention: a learned weight per neighbour; ours sums neighbours equally.
Training: the guide is an EMA at epoch 2,000 [15]; ours stop at 120.
Inputs: the oracle also reads nuclear-charge features.

Why we keep ours on our FM
Ours is disjoint **by construction**; TFG's only **by inference** — its saved arguments are overwritten by its own training script, and the oracle's training IDs are unknown. The price is a wider band (ours / TFG δ : 1.12× μ , 3.07× α , 2.00× gap), which is exactly why in-band never crosses backends.

Disjointness test (μ , 3,000 molecules): guide wins by >2× on 30.9%, oracle on 40.9%, neither on 28.2% — against 37.4 / 37.4 / 25.3% predicted for disjoint halves. Replicated on α and gap.

The data split, and why it has this shape

Split	Trains or supplies	n
train_a	generator (FM and VP) and guide f_A	51,527
train_b	evaluator f_B only	51,527
val	checkpoint selection; molecule sizes	17,748
test	never trained or selected on	13,083

δ calibration: 3,000 molecules from train_a + train_b; each of our nets is scored only on the half it did not train on.

QM9 [1], 133,885 molecules, pinned by SHA-256; split fixed by seed 20260917.

1 · f_A and f_B never share data

Guidance is an optimiser aimed at f_A; pushed hard it finds inputs that fool it. The checker needs different blind spots, so the guide-evaluator gap stays a usable reward-hacking signal.

2 · The generator may share with f_A

It judges nothing. This keeps the claim to one sentence: **f_B never saw a molecule the generator or the guide trained on.**

3 · Halves, as the literature does

EDM's conditional protocol and TFG's EDMsecond train on one ~50K half; TFG shipped a full-data EDMfull and *chose* EDMsecond [2, 15]. Using both halves would set twice the data beside every number we compare to. 70/30 was rejected for the same reason.

The cost we accept: half the data. A published half-data EDM loses only ~0.3 atom-stability points [2, 26] — so the split does not explain our gap in 2a.

The guidance arms, side by side

Shared frame. $m = x_t + (1 - t)v\theta \rightarrow$ frozen $f_A \rightarrow$ pull back through $J = \partial m / \partial x_t \rightarrow$ clip $\rightarrow$ add to the velocity ($t \geq 0.5, w = 1$).

Arm	Correction	What it changes	Passes / cell (2e)
DPS-style plug-in [12]	$G = \frac{y - f_A(m)}{s^2} J^\top \nabla f_A(m)$	the property at the endpoint; the baseline	10,000 / 4,000
TMPD-inspired [22]	$G = \frac{y - f_A(m)}{s^2 + g^\top \Sigma g} J^\top g, \quad \Sigma = k J, \quad g = \nabla f_A(m)$	shrinks the step where the endpoint is uncertain	10,000 / 4,000
LGD-MC [23]	$G = J^\top \sum_k \omega_k \frac{y - f_A(m + rz_k)}{s^2} \nabla f_A(m + rz_k), \quad r^2 = \frac{1}{d} \text{tr } \Sigma$	averages the likelihood over K draws around m	12,000 / 20,000
TFG [15]	$L(x) = \log \frac{1}{K} \sum_k \exp \left[- \left(\frac{f_A(x + \epsilon_k) - y}{\text{MAD}} \right)^2 \right]$ $\Delta_t = \rho \nabla_{x_t} L(m), \quad \Delta_0 : m \leftarrow m + \mu \nabla_m L(m) \quad (4 \text{ steps, no } J)$	adds steps on the predicted molecule itself	8,000 / 20,000
BDG (ours)	$\text{num}_i = (y - \bar{F}) - w_{\text{eff}} (F_i - \bar{F}), \quad w_{\text{eff}} = 1 + \eta e$	plug-in with a servoed deviation gain	10,000 / 4,000